Low-level EEG image reconstruction of our lab data: where we stand

Rapid-stream recordings E21-E29 (8 recordings, 5 people): 144 images at 5 per second, StarStim 32, no stimulation during the block. Target: the coarse 8x8 colour layout.
Score = coarse8: correlation of the reconstructed layout with the true one, relative to the other images of the recording (0 = chance, 1 = perfect).

THINGS-EEG2 benchmark
+0.44
10 people, 800 trials per test image

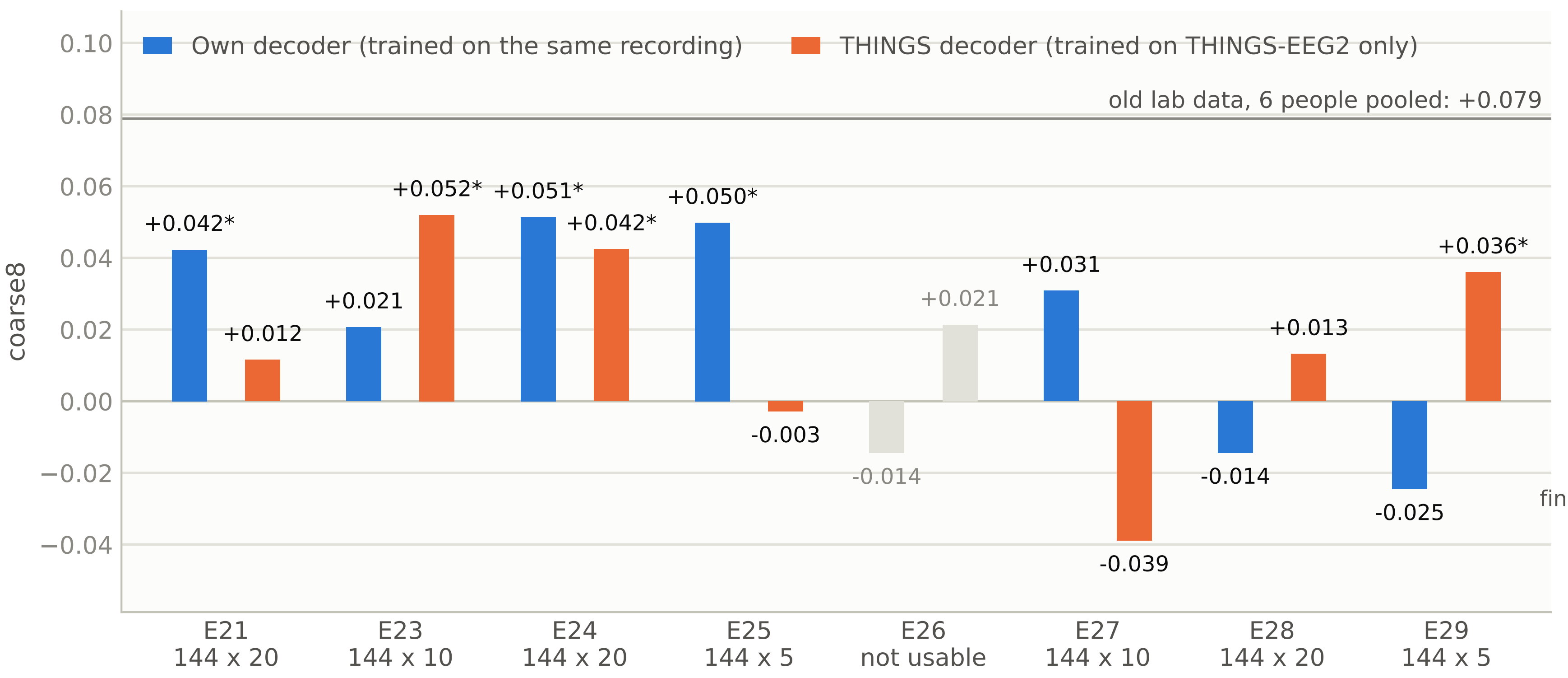
Old lab data, 6 people pooled
+0.079
single images, about 7 trials per image

Rapid stream, own decoder
+0.022
mean of 7 usable recordings; 3 above chance

Rapid stream, THINGS decoder
+0.016
mean of 7 usable recordings; 3 above chance

Reconstruction score per recording, corrected timing

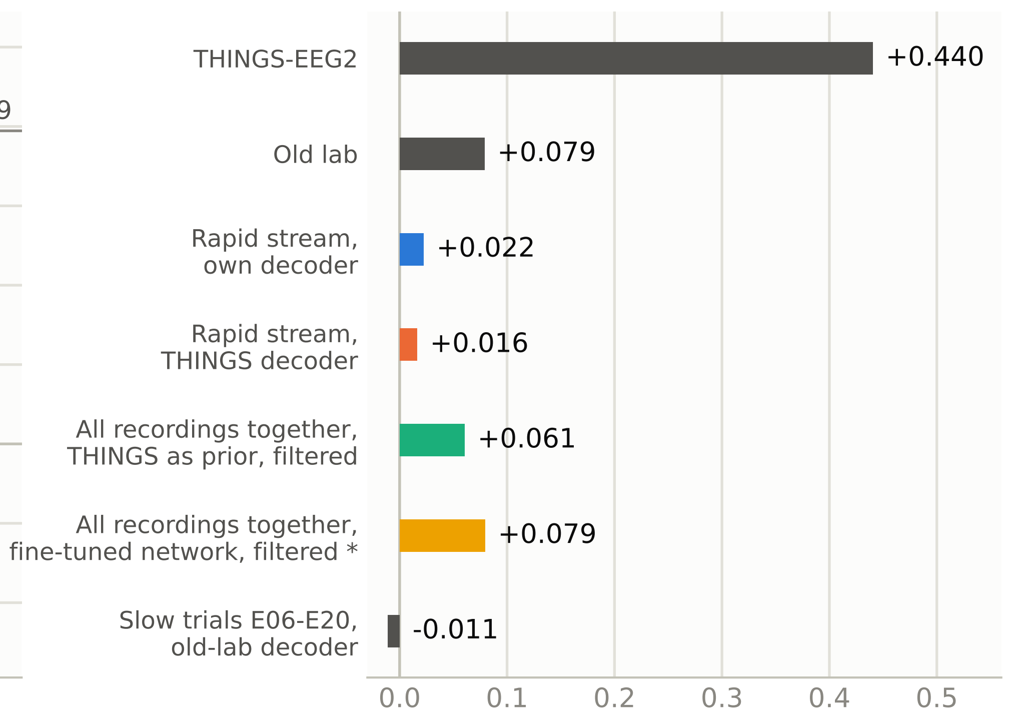
Two ways to decode the same recording. * = above chance (permutation test, $p < .05$).



Below each recording: images x repeats kept. Grey: timing could not be verified (E26).

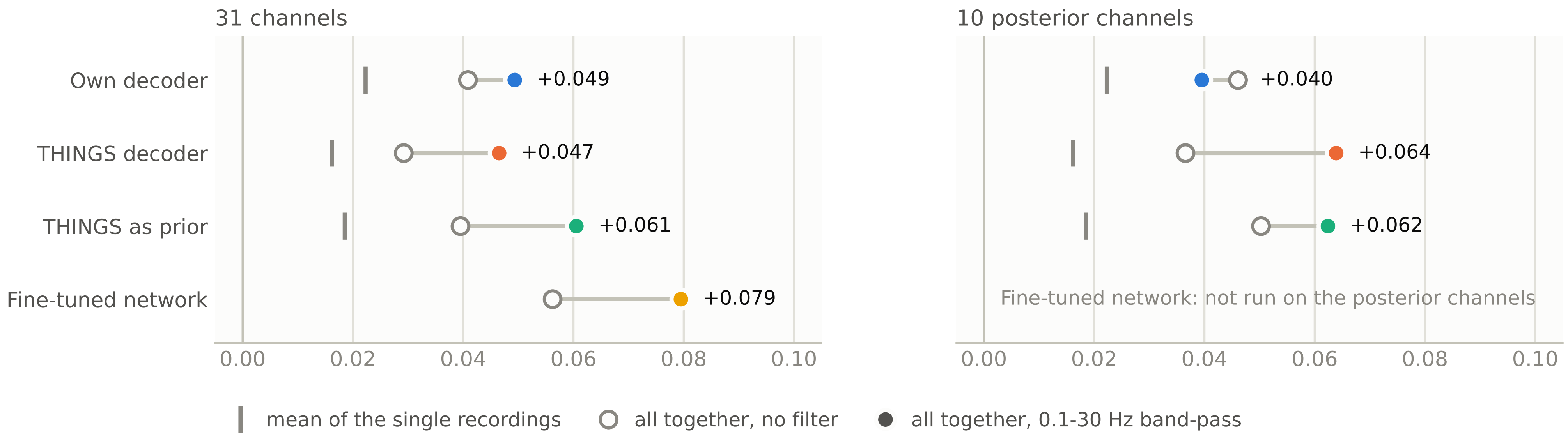
On the scale of the benchmark

The same score, all datasets.



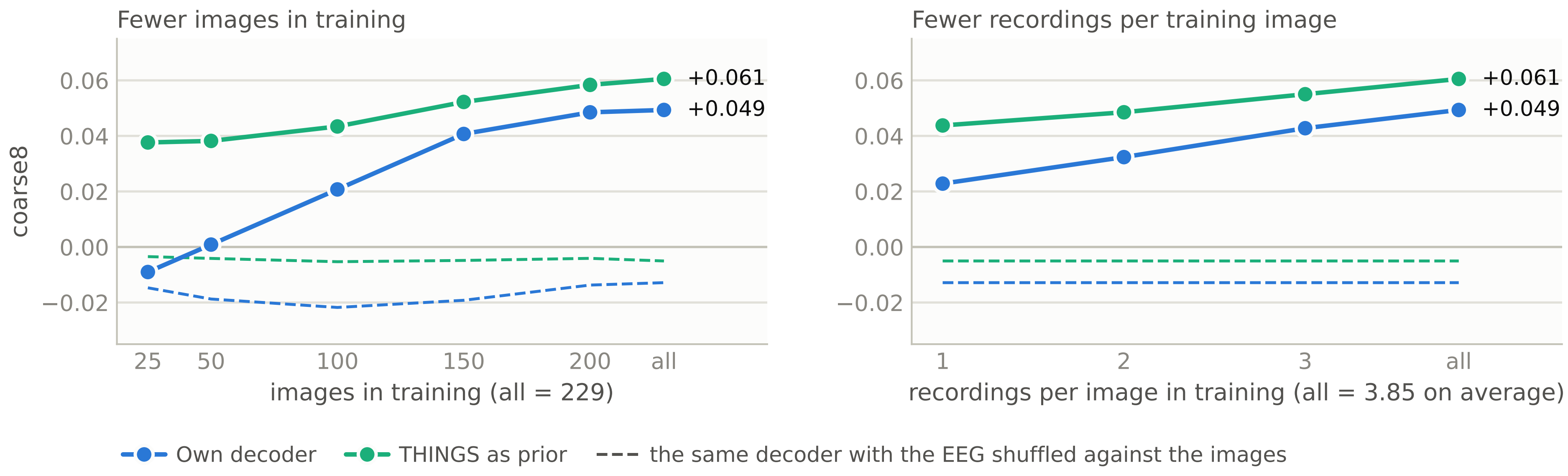
* pretrained on 173 of these 262 images; on the 80 it never saw: +0.061

All 7 usable recordings together: averaged per image, one decoder on 262 images



Fine-tuned network: the pixel network pretrained on THINGS-EEG2, all layers fine-tuned on our images, mean of 5 seeds; it has no single-recording value. It was pretrained on 173 of the 262 images, so its score is not free of leakage: on the 80 images it never saw it scores +0.079 without and +0.061 with the band-pass (own decoder on these: +0.048 and +0.057).

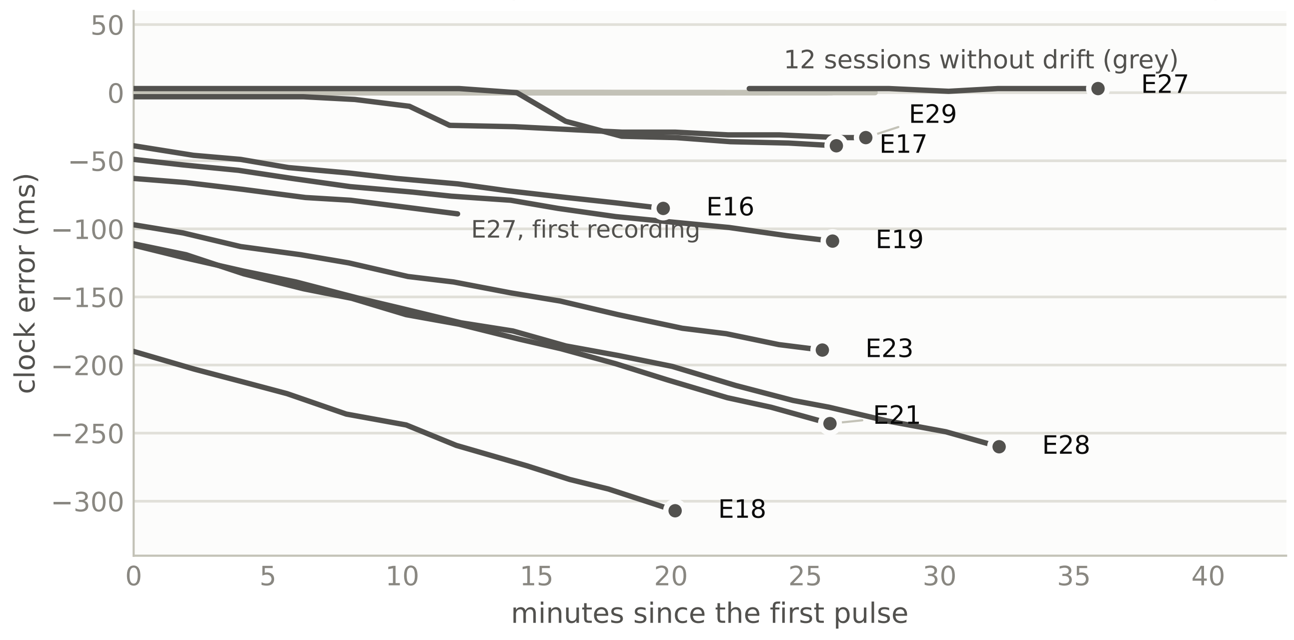
What more data would buy: the same decoders trained on less



0.1-30 Hz band-pass, 31 channels. The training images are thinned at random (mean of 20 draws); the test images stay at their full average. Both decoders still gain: from 100 to all images +0.029 (own decoder) and +0.017 (THINGS as prior), from 1 to all recordings +0.027 and +0.017. Scoring a test image from one recording instead of all costs 0.026 and 0.030. A decoder without signal scores below 0 (dashed).

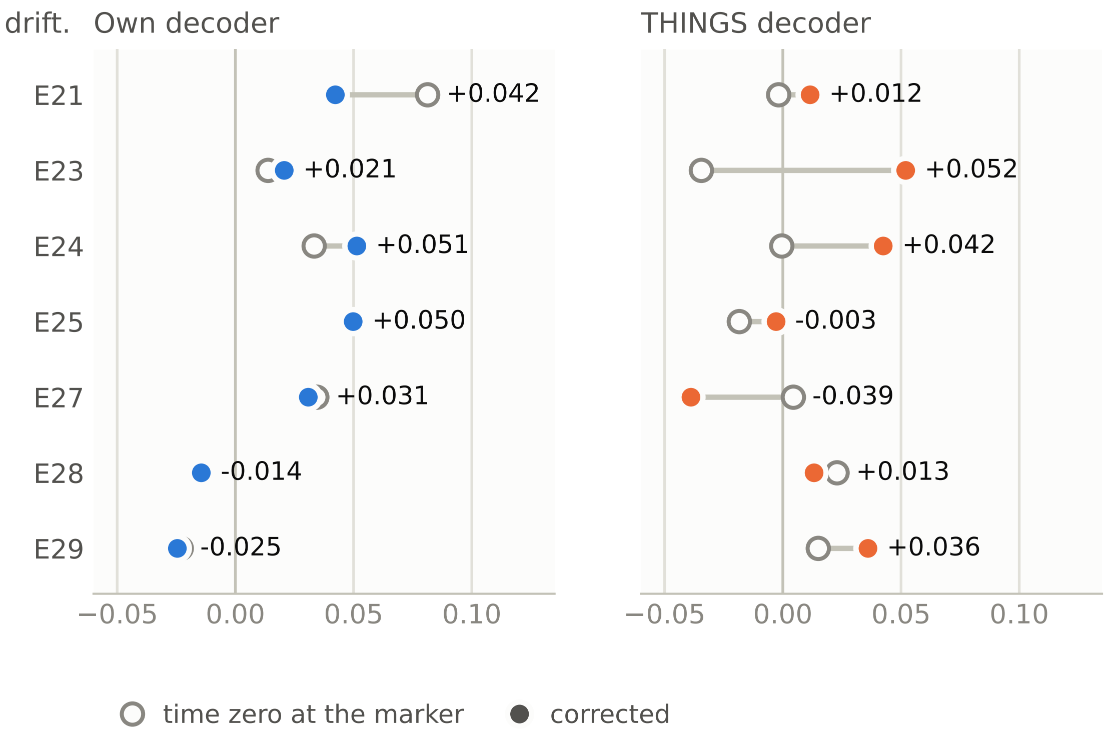
The EEG clock drifts in some sessions

Error of the EEG time stamps against the marker clock, from the stimulation pulses. 10 of 22 sessions drift.



E26: the other way round, EEG stamped +0.9 s to +2.1 s against the markers (not drawn).
Negative = an event shows in the EEG before its marker.

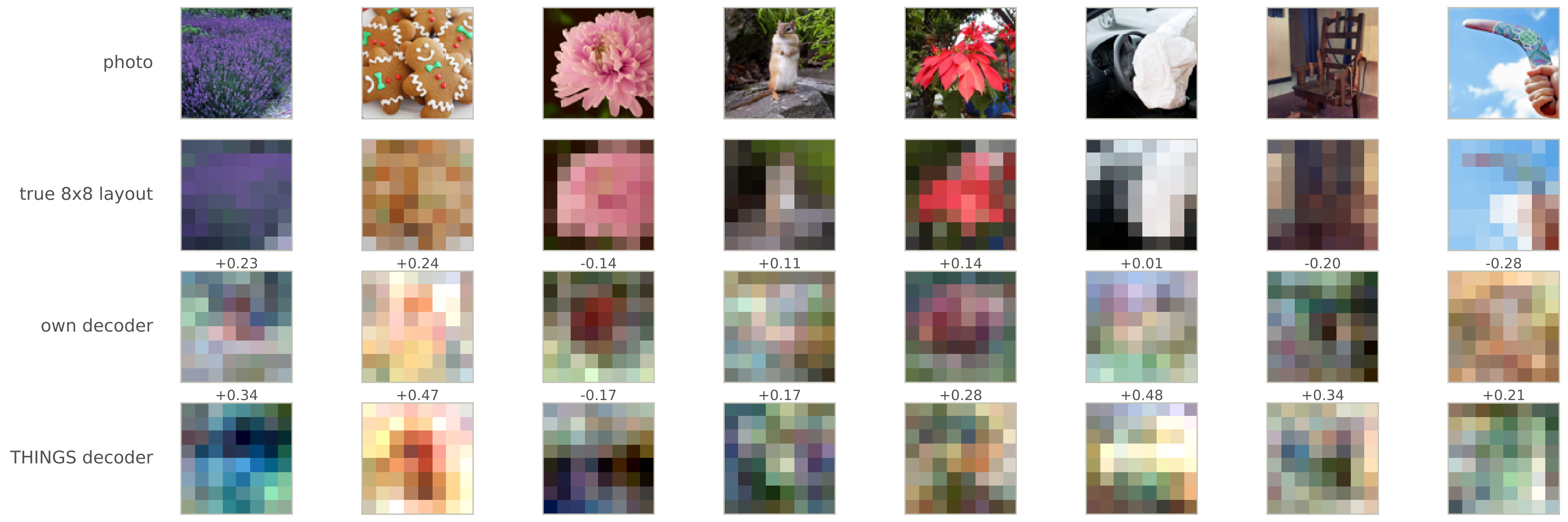
What the timing correction changes



Corrected = clock error removed, plus 46 ms for the delay of the image on our screen.

What the reconstructions look like: E24, 8 images drawn at random

Contrast-enhanced for display. The number above a reconstruction is the coarse8 of that image.



What is new

- 8 rapid-stream recordings from 5 people; 7 are usable, with 262 distinct images. Averaged per image and decoded together they reach +0.061 with a linear readout and +0.079 with a THINGS-pretrained network fine-tuned on them.
- The EEG clock runs off the marker clock in 10 of 22 sessions. Both loaders now measure this from the stimulation pulses and correct it, in the same way as memorability_stim.
- With corrected timing a decoder trained only on THINGS-EEG2 scores above chance in 3 of 7 recordings without any lab training data (best: E23, +0.052).

What still limits the reconstruction

- Together the recordings reach about 14% of the benchmark with a linear readout and 18% with the fine-tuned network (+0.06 and +0.08 against +0.44): 262 distinct images against 16,540 in THINGS-EEG2. Every recording shows one of two image sets, so new recordings add repeats, not images.
- The fine-tuned network is not reliably better than the linear readouts (+0.019, 95% interval -0.024 to +0.061, against THINGS as prior), and it was pretrained on 173 of the 262 images. On the 80 it never saw it scores +0.061, the own decoder +0.057.
- A single recording decodes weakly (+0.022 with its own decoder), and the old-lab decoder does not transfer to the slow image trials (-0.011).
- The timing is inferred from the data (pulses and images), not yet confirmed with a photodiode. E26 could not be used.